

## WTW 158 Worksheet 8 on Using Derivatives

### A. TEXTBOOK PROBLEMS

p283 no 27, 33, 39, 49, 55, 61

p 291 no 5, 13, 21

p 300 no 5, 9, 13a and b, 19, 29

### B. THEORY QUESTIONS

1. Complete the following definitions

(a) A critical number of a function  $f$  is a number  $c$  .....

(b) A function  $f$  is increasing on an interval  $I$  if .....

2. State the following theorems:

(a) The Extreme Value Theorem

(b) Fermat's Theorem

(c) The Mean Value Theorem

(d) The Increasing/Decreasing Test

(e) The Concavity Test

3. Write out the proof of the following theorem (The Decreasing/Increasing Test):

If  $f' > 0$  on an interval  $I$  then  $f$  is increasing on that interval.

### C. TEST QUESTIONS

1. Do the following functions have critical numbers in  $x = 0$ ?

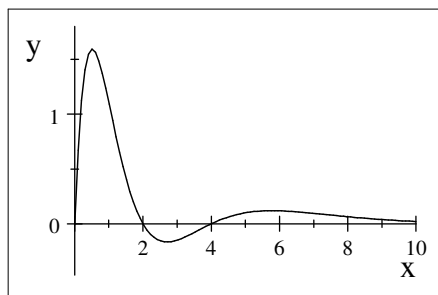
(i)  $f(x) = \sqrt{x}$  (ii)  $f(x) = \ln x$

2. Say for each of the following statements whether it is true or not. If not an example to illustrate why it is false.

(i) If  $f$  has a local extreme in  $x = c$  then  $f'(x) = 0$

(ii) If  $f'(x) = 0$  then  $f$  has a local extreme in  $x = c$ .

3. Consider the derivative graph of **the derivative function**  $f'$



(i) Where is  $f$  increasing? (ii) Where does  $f$  have a local maximum? (iii) Where is  $f$  concave up?

4. Consider  $f(x) = \frac{\ln x}{x}$

(i) Does the Closed Interval Method apply to the function  $f$  on  $[1, 6]$ ?

(ii) Find the absolute extremes on  $[1, 6]$ .

### D. SELECTED ANSWERS

A. See textbook.

B. 1., 2. and 3. See textbook.

C. 1. (a) Yes,  $x = 0$  is in the domain of the function and the derivative does not exist at  $x = 0$ . (b) No,  $x = 0$  is not in the domain of  $f$ .

2. (i) False,  $f$  has to be differentiable, eg  $f(x) = |x|$  in  $x = 0$ .

(ii) False, eg  $f(x) = x^3$  in  $x = 0$ .

3. (i)  $(0, 2) \cup (4, 10)$  (ii)  $x = 2$  (iii)  $(0, 1) \cup (3, 6)$

4. (i) Yes,  $f$  is continuous on  $[1, 6]$  (ii)  $f(1) = 0$  is the absolute minimum,  $f(e) = \frac{1}{e}$  (0, 37) is the local maximum.